

Project Report: Spring Data JPA - Quick Example

1. Project Overview

This project demonstrates how to integrate **Spring Boot**, **Spring Data JPA**, and **Hibernate** with a **MySQL** database to perform simple data persistence operations using the Country entity.

2. Software Pre-requisites

To run and build this project, the following software components are required:

- **MySQL Server 8.0**
- **MySQL Workbench 8**
- **Eclipse IDE for Enterprise Java Developers (2019-03 R)**
- **Apache Maven 3.6.2**
- **Java 8+**

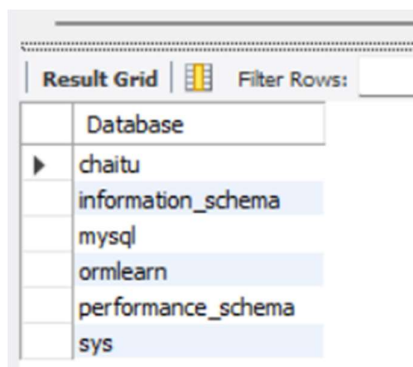
3. Project Setup Using Spring Initializr

- Navigate to <https://start.spring.io/>
- Configure the project as:
 - **Group:** com.cognizant
 - **Artifact:** orm-learn
 - **Description:** *Demo project for Spring Data JPA and Hibernate*
- Dependencies selected:
 - Spring Boot DevTools
 - Spring Data JPA
 - MySQL Driver
- Generate and download the .zip file
- Extract to Eclipse workspace
- Import using File > Import > Maven > Existing Maven Projects

4. MySQL Schema Setup

Create a new schema ormlearn:

```
mysql -u root -p
mysql> create schema ormlearn;
```



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Then create the table and insert data:

```
create table country(co_code varchar(2) primary key, co_name  
varchar(50));  
insert into country values ('IN', 'India');  
insert into country values ('US', 'United States of America');
```

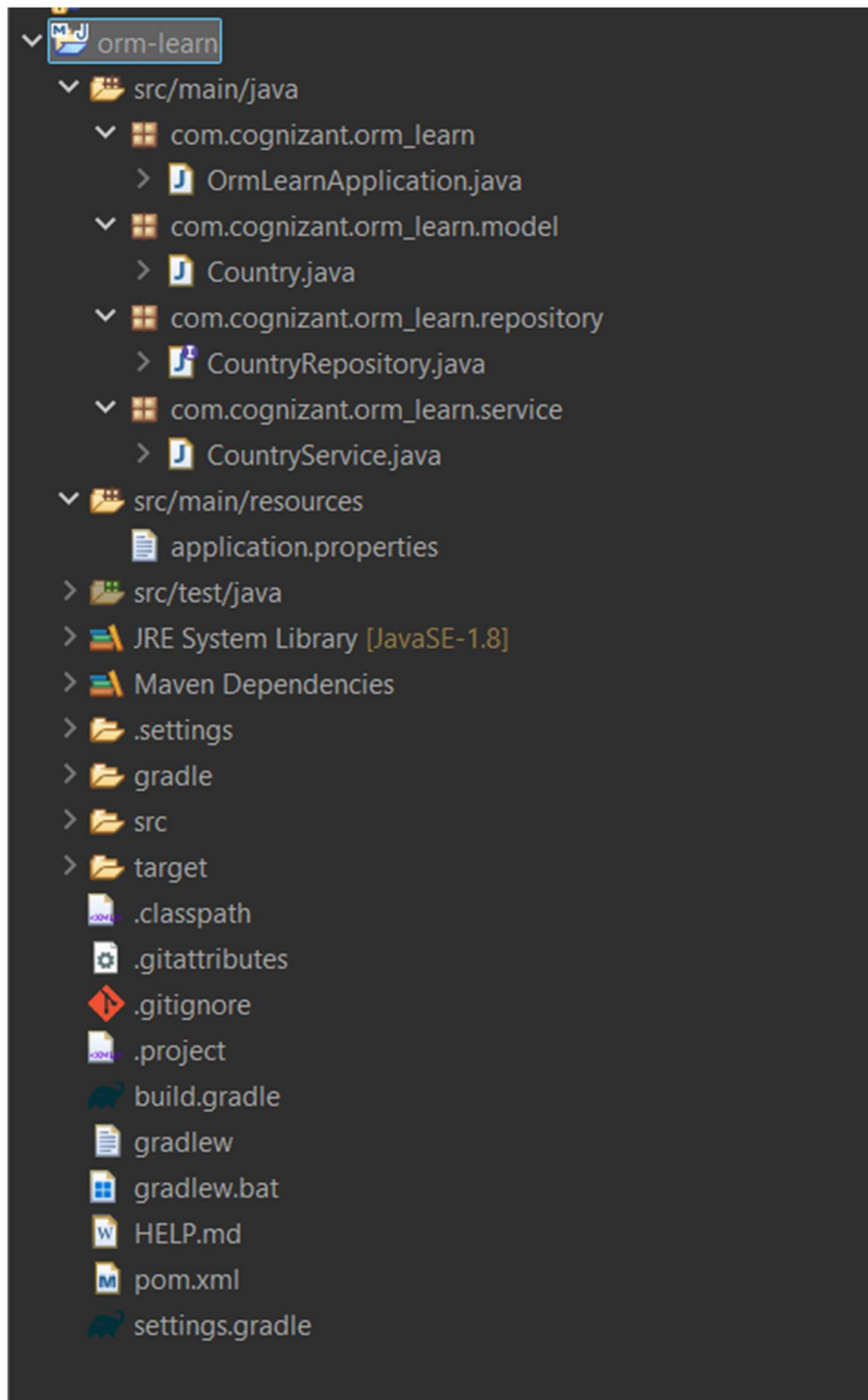
Result Grid  Filter Rows:

	Tables_in_ormlern
▶	country

Result Grid   Filter Rows:

	code	name
▶	IN	India
	US	United States of America
*	NULL	NULL

5. Project File Structure



6. application.properties Configuration

Location: src/main/resources/application.properties

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```
# Logging configuration
logging.level.org.springframework=info
logging.level.com.cognizant=debug
logging.level.org.hibernate.SQL=trace
logging.level.org.hibernate.type.descriptor.sql=trace

logging.pattern.console=%d{dd-MM-yy} %d{HH:mm:ss.SSS} %-20.20thread
%5p %-25.25logger{25} %25M %4L %m%n

# Database configuration
spring.datasource.driver-class-name=com.mysql.cj.jdbc.Driver
spring.datasource.url=jdbc:mysql://localhost:3306/ormlearn
spring.datasource.username=root
spring.datasource.password=root

# Hibernate configuration
spring.jpa.hibernate.ddl-auto=validate
spring.jpa.properties.hibernate.dialect=org.hibernate.dialect.MySQL
5Dialect
```

8. Java Class Summary

8.1 Entity Class: Country

Package: com.cognizant.ormlearn.model

```
package com.cognizant.orm_learn.model;

import javax.persistence.Column;
import javax.persistence.Entity;
import javax.persistence.Id;
import javax.persistence.Table;

@Entity
@Table(name = "country")
public class Country {

    @Id
    @Column(name = "code")
    private String code;

    @Column(name = "name")
    private String name;

    public String getCode() {
        return code;
    }

    public void setCode(String code) {
        this.code = code;
    }

    public String getName() {
        return name;
    }

    public void setName(String name) {
```

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```
        this.name = name;
    }

    @Override
    public String toString() {
        return "Country [code=" + code + ", name=" + name + "]";
    }
}
```

8.2 Repository Interface: CountryRepository

Package: com.cognizant.ormlearn.repository

```
package com.cognizant.orm_learn.repository;

import org.springframework.data.jpa.repository.JpaRepository;
import com.cognizant.orm_learn.model.Country;

public interface CountryRepository extends JpaRepository<Country,
String> {
}
```

8.3 Service Class: CountryService

Package: com.cognizant.ormlearn.service

```
package com.cognizant.orm_learn.service;

import java.util.List;
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.stereotype.Service;
import com.cognizant.orm_learn.model.Country;
import com.cognizant.orm_learn.repository.CountryRepository;

@Service
public class CountryService {

    @Autowired
    private CountryRepository countryRepository;

    public List<Country> getAllCountries() {
        return countryRepository.findAll();
    }
}
```

8.4 Main Class: OrmLearnApplication.java

Package: com.cognizant.ormlearn

```
package com.cognizant.orm_learn;

import java.util.List;

import org.slf4j.Logger;
import org.slf4j.LoggerFactory;
import org.springframework.boot.SpringApplication;
```

```
import
org.springframework.boot.autoconfigure.SpringBootApplication;
import org.springframework.context.ApplicationContext;
import
org.springframework.data.jpa.repository.config.EnableJpaRepositories;
import org.springframework.boot.autoconfigure.domain.EntityScan;
import org.springframework.context.annotation.ComponentScan;
import com.cognizant.orm_learn.model.Country;
import com.cognizant.orm_learn.service.CountryService;

@SpringBootApplication
public class OrmLearnApplication {

    private static final Logger LOGGER =
LoggerFactory.getLogger(OrmLearnApplication.class);

    private static CountryService countryService;

    public static void main(String[] args) {
        ApplicationContext context =
SpringApplication.run(OrmLearnApplication.class, args);
        LOGGER.info("Inside main");

        countryService = context.getBean(CountryService.class);
        testGetAllCountries();
    }

    private static void testGetAllCountries() {
        LOGGER.info("Start");
        List<Country> countries = countryService.getAllCountries();
        LOGGER.debug("countries={}", countries);
        LOGGER.info("End");
    }
}
9. Important Annotations Used
```

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Output Logs (Console)

```

<terminated> OmlearnApplication [Java Application] [C:\Program Files\Java\jre1.8.0_451\bin\javaw.exe (07-Jul-2025, 9:59:48am - 9:59:51am elapsed: 00:00:03.800) [pid: 13224]

Console X
=====
      _ _ _ _ _
     / / / / /
    / / / / /
   / / / / /
  / / / / /
 / / / / /
/ / / / /
=====
:: Spring Boot ::

(v2.7.7)

07-07-25 09:59:49.044 main INFO c.c.o.OmlearnApplication
07-07-25 09:59:49.046 main DEBUG c.c.o.OmlearnApplication
07-07-25 09:59:49.047 main INFO c.c.o.OmlearnApplication
07-07-25 09:59:49.349 main INFO totyconfigurationDelegate
07-07-25 09:59:49.399 main INFO totyconfigurationDelegate
07-07-25 09:59:49.831 main INFO o.h.j.i.util.LogHelper
07-07-25 09:59:49.864 main INFO org.hibernate.Version
07-07-25 09:59:49.988 main INFO o.h.a.common.Version
07-07-25 09:59:50.211 main INFO c.z.h.HikariDataSource
07-07-25 09:59:50.717 main INFO c.z.h.HikariDataSource
07-07-25 09:59:50.730 main INFO o.h.dialect.Dialect
07-07-25 09:59:51.271 main INFO p.i.JpaPlatformInitiator
07-07-25 09:59:51.278 main INFO EntityManagerFactoryBean buildEntityManagerFactory
07-07-25 09:59:51.695 main INFO c.c.o.OmlearnApplication
07-07-25 09:59:51.701 main INFO c.c.o.OmlearnApplication
07-07-25 09:59:51.701 main INFO c.c.o.OmlearnApplication
07-07-25 09:59:51.839 main DEBUG org.hibernate.SQL
07-07-25 09:59:51.850 main TRACE o.h.t.d.s.BasicExtractor
07-07-25 09:59:51.857 main TRACE o.h.t.d.s.BasicExtractor
07-07-25 09:59:51.858 main TRACE o.h.t.d.s.BasicExtractor
07-07-25 09:59:51.859 main TRACE o.h.t.d.s.BasicExtractor
07-07-25 09:59:51.871 main DEBUG c.c.o.OmlearnApplication
07-07-25 09:59:51.871 main INFO c.c.o.OmlearnApplication
07-07-25 09:59:51.877 main INFO EntityManagerFactoryBean
07-07-25 09:59:51.877 main INFO c.z.h.HikariDataSource
07-07-25 09:59:51.887 main INFO c.z.h.HikariDataSource

LogStarting 55 Starting OmlearnApplication using Java 1.8.0.451 on hp with PID 13224 (C:\Users\hnuh\workspace\work
LogStarting 56 Running with Spring Boot v2.7.7, Spring v5.3.24
LogStartingProfileInfo 631 No active profile set, falling back to 1 default profile: "default"
registerRepositoryIn 132 Bootstrapping Spring Data JPA repositories in DEFAULT mode.
registerRepositoryIn 201 Finished Spring Data repository scanning in 36 ms. Found 1 JPA repository interfaces.
LogPersistenceUnitInformation 31 HH0000204: Processing PersistenceUnitInfo [name: default]
LogVersion 44 HH0000412: Hibernate ORM core version 5.6.14.Final
<clinit> 56 HCANW000001: Hibernate Commons Annotations {5.1.2.Final}
getConnection 110 HikariPool-1 - Starting...
getConnection 123 HikariPool-1 - Start completed.
<clinit> 175 HH0000400: Using dialect: org.hibernate.dialect.MySQL5Dialect
initiateService 52 HH0000490: Using JpaPlatform implementation: [org.hibernate.engine.transaction.jta.platform.internal
LogStarted 61 Started OmlearnApplication in 2.986 seconds (JVM running for 3.385)
main 30 Inside main
37 Start
testGetAllCountries 144 select country0_.code as code1_0_, country0_.name as name2_0_, from country country0_
extract 60 extracted value ((code1_0_) : [VARCHAR]) - [IN]
extract 60 extracted value ((name2_0_) : [VARCHAR]) - [India]
extract 60 extracted value ((code1_0_) : [VARCHAR]) - [US]
extract 60 extracted value ((name2_0_) : [VARCHAR]) - [United States of America]
testGetAllCountries 39 countries=[Country [code=IN, name=India], Country [code=US, name=United States of America]]
40 End
destroy 651 Closing JPA EntityManagerFactory for persistence unit 'default'
close 350 HikariPool-1 - Shutdown initiated...
close 352 HikariPool-1 - Shutdown completed.
```

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Conclusion

This quick-start project provides a strong foundation for implementing CRUD operations using **Spring Boot**, **JPA**, and **Hibernate**. It showcases how minimal configuration enables rapid development and clean abstraction using the repository and service layers.